

Lecture 15 - Economics and Business Administration

Cost Management

Agenda

- Cost Classification
- Direct and Indirect Costs
- Product Costs and Period Costs
- Cost Allocation

Cost Classification

Classifications relevant to decision-making processes (relationships between costs and cost objects):

- ✓**direct costs;**
- ✓**indirect costs;**
- ✓**unallocated costs**

Direct and Indirect Costs

Costs must be allocated to a variety of cost objects for different purposes:

- verify the selling price of a product or a job order;
- calculate the cost of a production department to decide on outsourcing;
- determine the profitability of a product line;
- …

The distinction between direct and indirect relates to how costs are traced to the cost object:

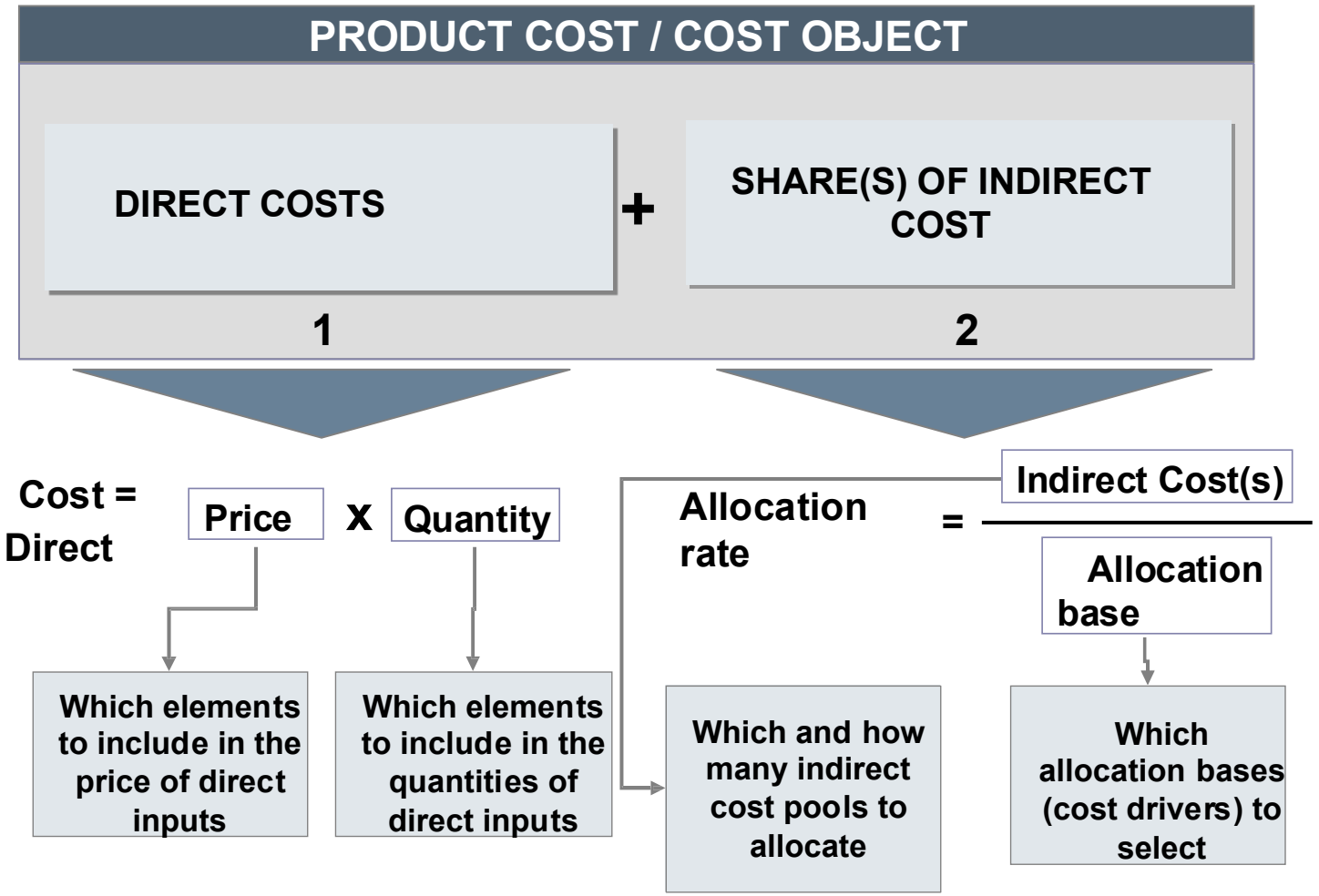
Direct Costs

- Can be traced to a unit of product or another cost object in a specific and exclusive manner
- Example: direct materials used in the production of a product; commissions paid to salespeople on

Indirect Costs

- Cannot be traced to a unit of product or other cost object in a specific and exclusive manner.
- However, they can be assigned to cost objects through an allocation process using reliable output measures (cost drivers).
- Example: depreciation of a production plant

Product Cost Determination



Production Costs

Manufacturing companies organise production costs into 3 broad categories:

✓**Direct Materials (DM):** materials that become an integral part of the product and can be easily traced to it (e.g. car tyres)

✓**Direct Labour (DL):** labour costs that can be easily traced to individual units of product (e.g. wages paid to workers on a car production line)

✓**Manufacturing Overhead**

DM Cost	**+**	**DL Cost**	**+**	**Manufacturing Overhead**	**=**	**Full Manufacturing Cost**

Manufacturing Overhead

Production costs that **cannot be directly traced to specific units produced** (individual product)

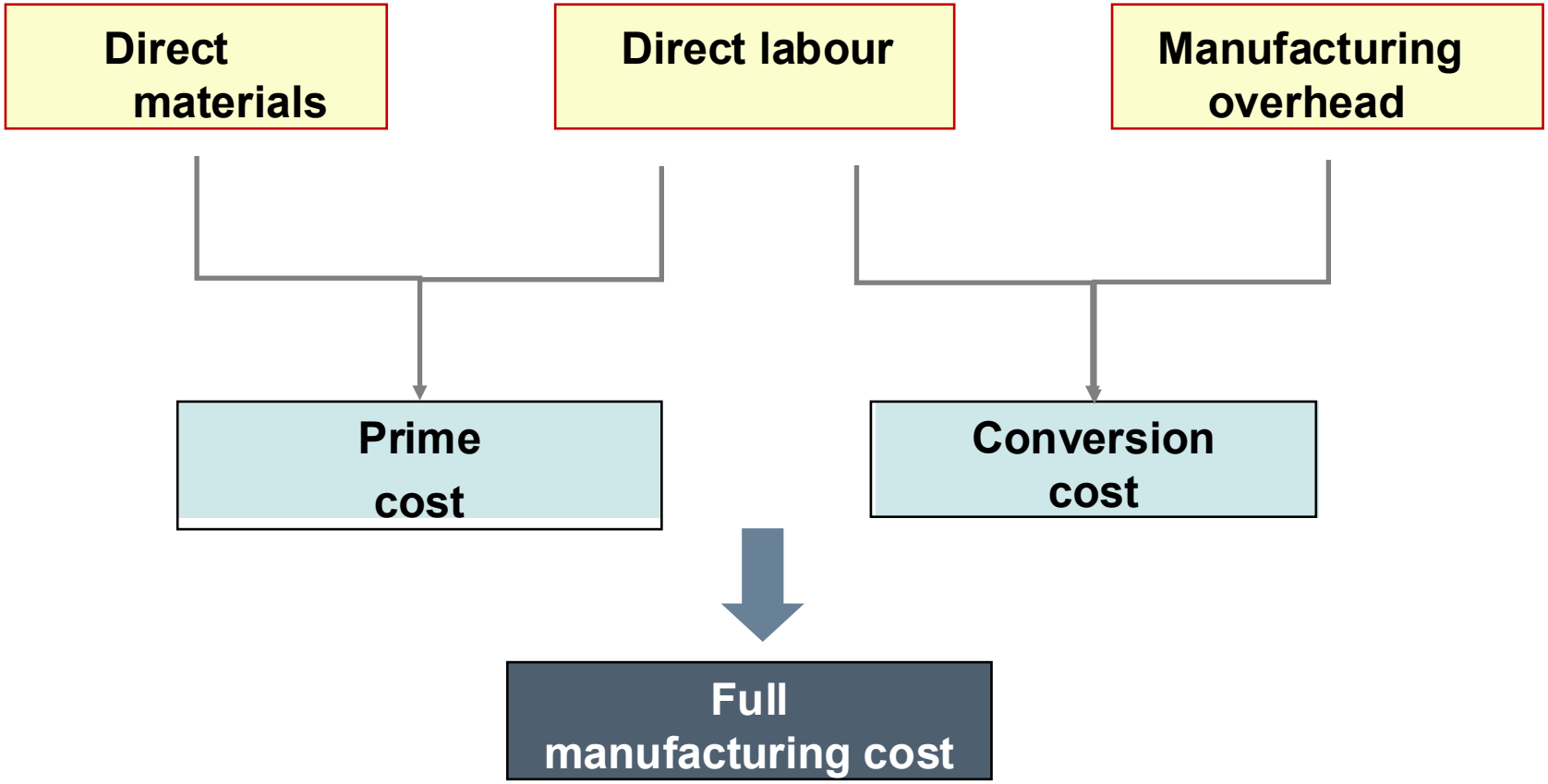
✓**Indirect Labour** → Wages paid to employees not directly involved in production work. E.g. *maintenance staff, production schedulers, janitors and security guards, etc.*

✓**Indirect Materials** → Auxiliary materials in the production process. E.g. *lubricants and cleaning supplies used in the car assembly plant*

Cost measurement systems establish the methods by which these costs can be measured (indirectly) with reference to the individual product, in order to determine its full manufacturing cost

Classification of Production Costs

Production costs are often classified as follows



Non-Production Costs: Examples

Marketing and selling costs

Costs required to secure orders and bring the product to market

Administrative and general costs

Costs related to the administrative and management structures of the company

Unallocated Costs

- Costs for which **no relationship with a cost object can be identified** and which, for reasons of practicality or reliability, it is preferable not to allocate.
- Examples: *research and development, accounting and control, legal expenses, information systems, etc.*

Note:

- All costs could theoretically be allocated using simple drivers (e.g. *revenues or production volumes*). However, to preserve the relationship between allocated costs and resource use intensity, where **such drivers are not appropriate it is preferable not to allocate costs.**
- Unallocated costs vary depending on the type of company, value chain, and operating systems. Costs that are unallocated in one company may be indirect costs in another.

Income Statement of a Manufacturing Company

Revenues

Revenue adjustments
-Commissions
-Returns
-Allowances

Net revenues

Production costs

Gross Margin

Non-production costs

Operating Income

Also referred to as:
- ✓Gross Profit
- ✓Gross Margin

… For a Manufacturing Company

NON-PRODUCTION costs (general overhead costs)

Costs incurred for the general operation of the company

E.g. costs related to the administrative office, selling and commercial costs, marketing costs, general management costs, etc.

PRODUCTION costs

Costs incurred for activities that directly support production

E.g. costs related to the technical office, production scheduling, raw materials warehouse, quality control unit, etc.

Costs incurred for actual production (transformation of raw materials into finished goods)

E.g. costs of raw materials used, workers in the factory, depreciation of production machinery, electricity consumed, etc.

Production Costs and the Accruals Principle

- The accruals principle, as covered in the financial accounting course, requires that costs be “expensed” in the income statement consistently with the revenues they generate.
- Costs are in fact incurred mainly in relation to the purchase and use of production inputs (and only in rare cases in relation to the sale: e.g. commissions or selling costs)
- Where production volumes and sales volumes differ, the accruals principle is applied through inventory valuation:

- **Raw materials inventory:** valuation of purchases not yet drawn for production
- **Finished goods inventory:** valuation of goods produced but not yet sold
- **Work-in-progress inventory:** valuation of products not yet completed

Cost Measurement Systems (CMS) provide the information needed for inventory valuation (finished goods and work-in-progress)

Full Manufacturing Cost

- As is known, the Civil Code and national accounting standards require that **inventory valuation** includes **all costs incurred**.
- Historical cost is defined by national accounting standards as the "**total costs incurred to obtain ownership of inventory in its current location and condition**" (Accounting Standard no. 11). Historical cost therefore corresponds to:

➢ the **purchase cost** for **raw materials, auxiliary materials, consumables, and purchased semi-finished goods**;

➢ the **production (or manufacturing) cost** for **finished goods, manufactured semi-finished goods, and work-in-progress**

Cost measurement systems - when applied to measuring the cost of products manufactured by the company (cost object: individual product) - define the methods by which the company determines the unit value of production costs.
The **unit production cost** constitutes the Full Manufacturing Cost (full industrial cost, full production cost)



Product Costs and Period Costs

Product Costs and Period Costs

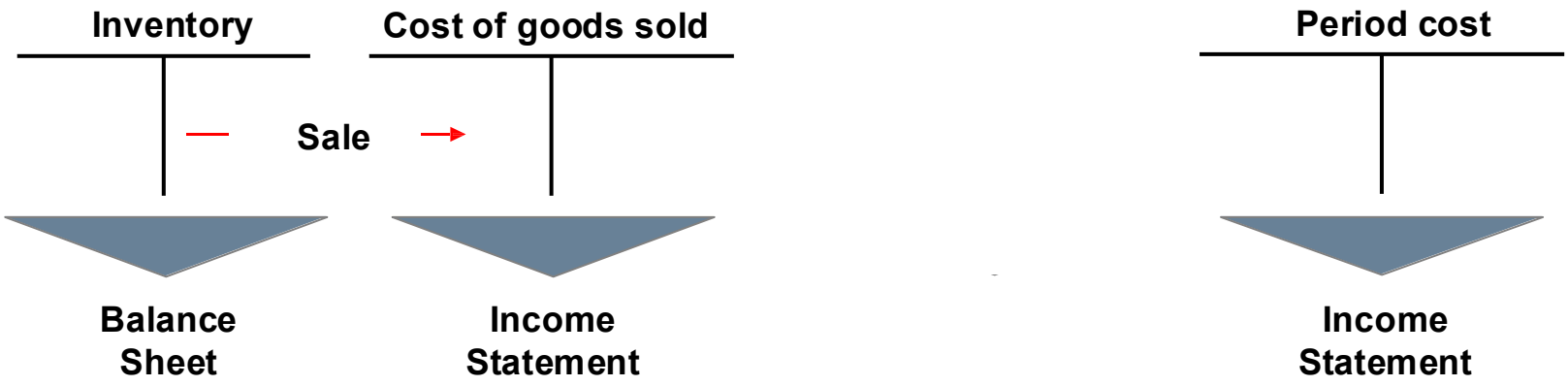
This distinction is driven by the accruals principle, under which costs incurred to generate a given revenue should be “recognised” in the same period in which the revenue is earned.

Product Costs

- are costs assigned to the product
- contribute to defining the value of inventories (finished goods and semi-finished goods)
- form the Cost of Goods Sold
- include direct materials, direct labour, and manufacturing overhead (full manufacturing cost).

Period Costs

- Are not included in product costs
- Do not contribute to the value of period-end inventories (finished goods and semi-finished goods)
- Are fully expensed in the Income Statement
- Refer to non-production costs



Cost of Goods Sold

In this framework, the Cost of Goods Sold can be defined as the FULL MANUFACTURING COST of products that have been sold, which must therefore be "expensed" in the income statement on an accruals basis against revenues

- ✓ Total production costs = €100,000
- ✓ Non-production costs = €20,000
- ✓ Number of units produced = 1,000
- ✓ Unit full manufacturing cost = €100
- ✓ Number of units sold = 900
- ✓ Cost of Goods Sold = €100 × 900 = €90,000
- ✓ Increase in finished goods inventory = 100 units × €100 = 10,000

Not all manufacturing costs incurred will therefore be charged to the income statement: €10,000 of such costs will be reversed at inventory valuation and will accordingly be deferred on the balance sheet.

Conversely, all €20,000 of non-production costs, being period costs, will be fully charged to the income statement

Cost of Goods Sold Structure: An Example

Trading Company

Opening merchandise inventory	**45**
Merchandise purchases	**140**
Closing merchandise inventory	**-125**
Cost of goods sold	**60**

Income Statement

Revenues	100
Cost of goods sold	60
Gross Margin	40
Selling, General and Administrative (SG&A) costs	30
Operating Income	10

Manufacturing Company

Opening raw materials inventory	**45**
Raw materials purchases	**80**
Closing raw materials inventory	**-100**
Raw materials consumed	**25**

Manufacturing Company

Raw materials consumed	25
Direct labour	40
Manufacturing overhead	25
Manufacturing depreciation	15
Factory supervision	10
Opening WIP inventory	45
Closing WIP inventory	-60
Cost of goods manufactured	100
Opening finished goods inventory	190
Closing finished goods inventory	-230
Change in finished goods inventory	-40

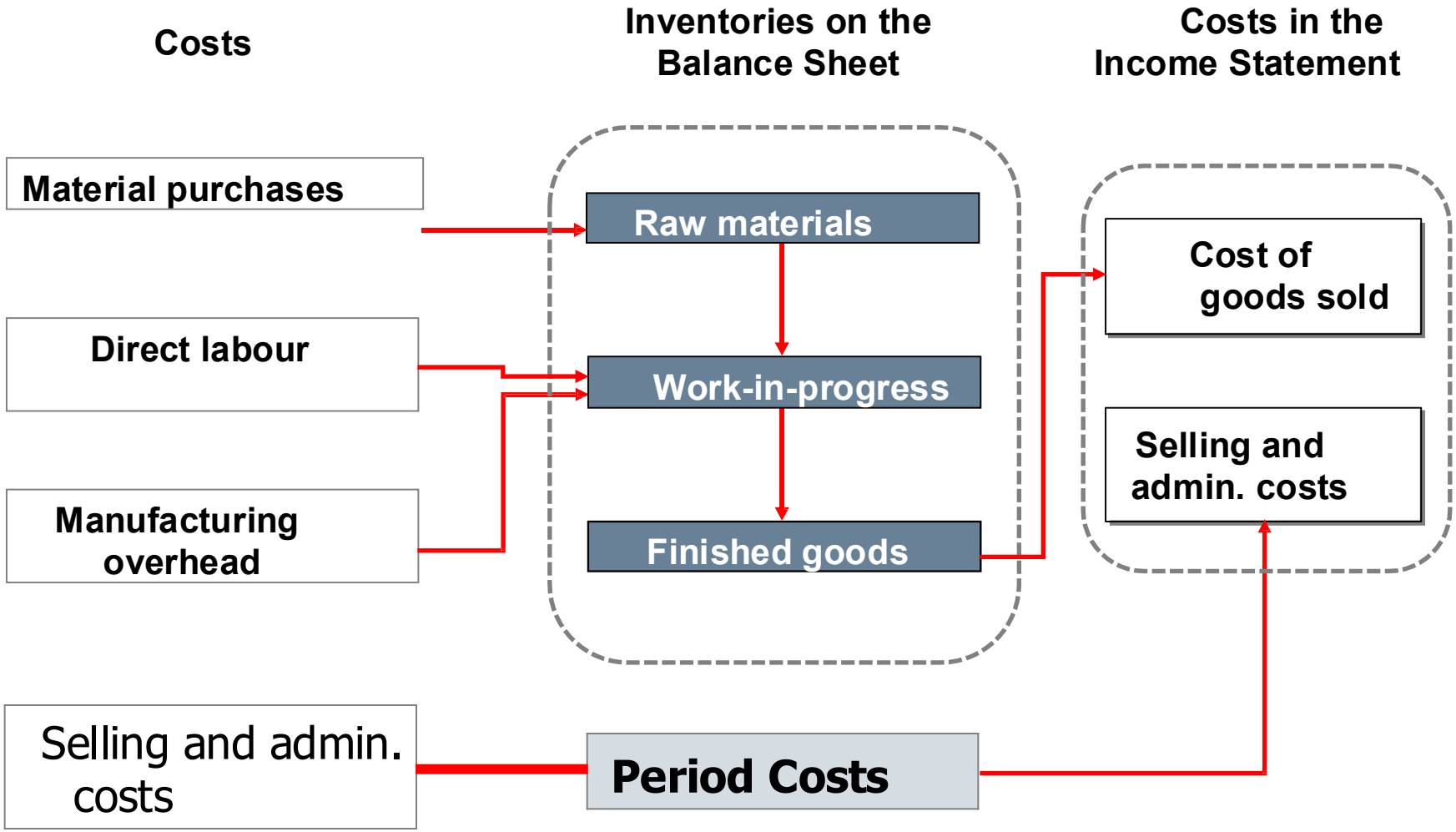
Change in WIP inventory: -15

Cost of Goods Sold: 60

Product Cost Structure (Finished Goods / WIP)

Raw materials consumed	Direct labour	Manufacturing overhead

Flow of Production Costs





Cost Allocation

Cost Assignment and Allocation

Cost Assignment

The process that **"traces"** costs to an **object** at the time they are recorded. For example, a cost incurred with direct and exclusive reference to a department is "assigned" to it in a specific manner.

Cost Allocation and Apportionment

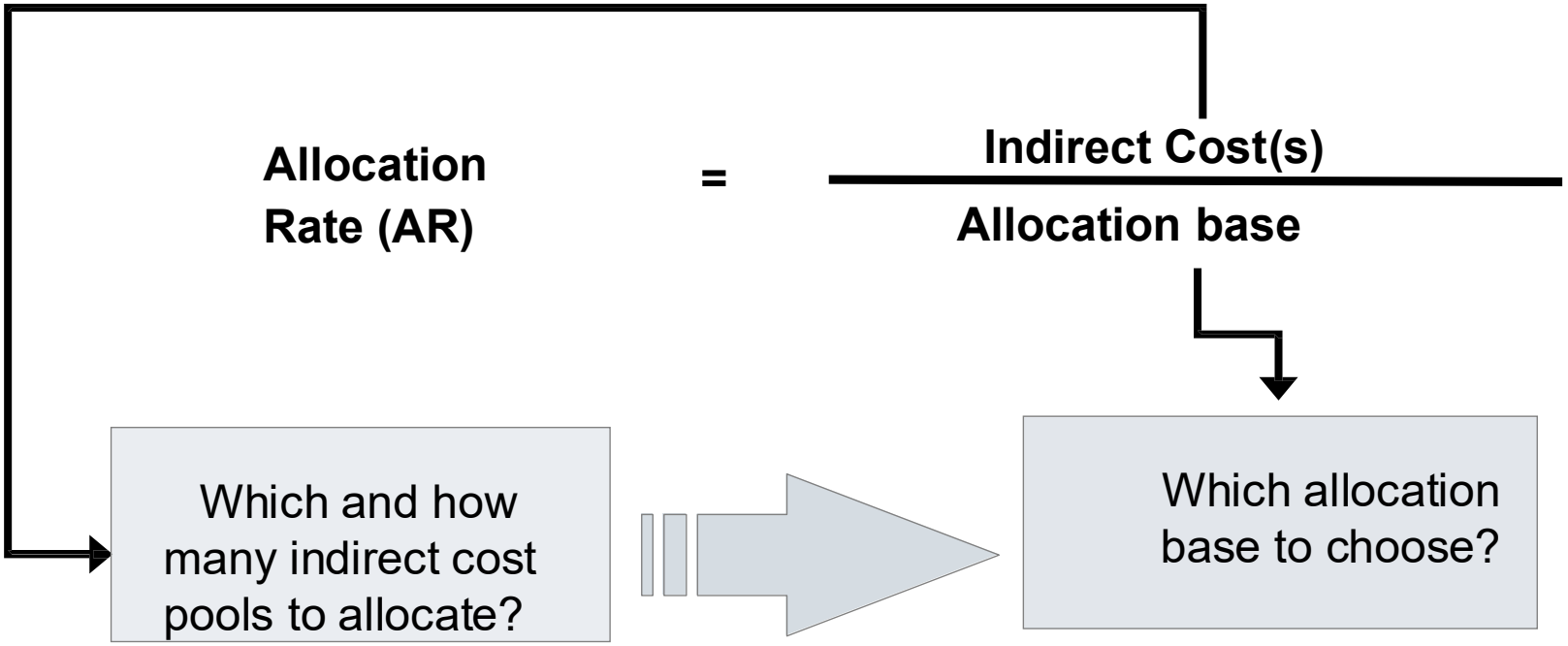
The subsequent process by which a cost, previously "assigned" to one object (e.g. maintenance department), is **referred to a different object** (e.g. products A and B that benefit from that department's maintenance).

Indirect Cost Allocation

- The allocation of **Indirect costs** is a significant challenge in determining the cost of cost objects (and products in particular).
- Typically, an increase in investment intensity (degree of automation) associated with production leads to a higher share of indirect costs.

- There is **no indirect cost allocation method** that can yield a value definable as the true cost of the product.
- All indirect cost allocation methods are based on **assumptions, estimates, allocations and calculations**.
- It is the responsibility of the Management Accounting officer (controller) to select cost determination methodologies that are compatible:
  - ✓ with the **characteristics of the production system,**
  - ✓ with the **information needs of management** (relevance of cost information).

Indirect Cost Allocation Criterion





Cost Allocation: Base Example

Cost Allocation Exercise

Manufacturing overhead = €1,000,000

2 products: A and B

Volumes:

- A = 2.000
- B = 3.000

Direct labour hours used

- A = 100
- B = 300

Machine hours used

- A = 500
- B = 500

- Direct materials consumed
  - A = € 800.000
  - B = € 200.000

Allocate manufacturing overhead to products A and B

Cost Allocation Exercise: Volume-Based Allocation

Costs to allocate	€ 1.000.000
Allocation base	1) Volumes:

$$\text{Allocation Rate (AR)} = \frac{\text{€ 1.000.000}}{\text{(2.000+3.000)}} = 200$$

Costs allocated to each product

A = 200 x 2.000 → 400.000

B = 200 x 3.000 → 600.000

Cost Allocation Exercise: Direct Labour Hours Allocation

Costs to allocate	€ 1.000.000
Allocation base	2) DL hours:

Allocation Rate (AR) = $\frac{\text{€ 1.000.000}}{(100+300)}$ = 2.500

Costs allocated to each product

- A = 2.500 x 100 → 250.000
- B = 2.500 x 300 → 750.000

Cost Allocation Exercise: Machine Hours Allocation

Costs to allocate	**€ 1.000.000**
Allocation base	**3) Machine hours:**

Allocation Rate (AR) = $\frac{\text{€ 1.000.000}}{(500+500)}$ = 1.000

Costs allocated to each product

A = 1.000 x 500 → 500.000

B = 1.000 x 500 → 500.000

Cost Allocation Exercise: Direct Materials Consumed Allocation

Costs to allocate	€ 1.000.000

Allocation base	4) Direct Materials consumed:

Allocation Rate (AR) = € 1.000.000 / (800.000 + 200.000) = 1

Costs allocated to each product

A = 1 x 800.000 → 800.000

B = 1 x 200.000 → 200.000